

Notes: Benmelech and Bergman (JF, 2011)
Bankruptcy and the Collateral Channel

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Research question and hypothesis	4
4	Institutional setting: airline secured debt	4
5	Data and construction of key variables	4
5.1	Bankrupt Buyers	5
5.2	Bankrupt Aircraft	5
5.3	Redeployability	5
6	Identification strategy	5
7	Main empirical findings	6
7.1	Bankruptcy shocks raise spreads	6
7.2	Effects are stronger for exposed securities	6
7.3	Effects are temporary and tied to bankruptcy periods	6
7.4	Robustness checks reject alternative channels	6
8	Conceptual takeaway	7
9	Potential limitations and useful cautions	7
10	Extracted figures and tables from the paper	8
10.1	Table I: EETC Structures	8
10.2	Table II: Summary Statistics	8
10.3	Figure 1: Total number of bankrupt potential buyers for Boeing 737 and Boeing 747	9
10.4	Figure 2: Total number of Boeing 737 and Boeing 747 aircraft operated by bankrupt airlines in the United States	9
10.5	Table III: Bankrupt Buyers and Bankrupt Aircraft Measures	9

10.6 Table IV: Bankruptcy, Bankrupt Buyers, and Tranche Credit Spreads: Univariate Analysis	10
10.7 Table V: Bankruptcy and Collateral	10
10.8 Table VI: Bankruptcy and Collateral: Credit Spreads of Nonbankrupt Airlines . . .	11
10.9 Table VII: The Time Series of the Collateral Channel	11
10.10Table VIII: The Collateral Channel and Industry Conditions	12
10.11Table IX: Bankruptcy, Collateral, Tranche Seniority, and LTV	12
10.12Table X: The Collateral Channel and Airlines' Financial Strength	13
10.13Table XI: The Collateral Channel and Tranche Fleet	13
10.14Table XII: Bankruptcy Collateral and Investment Grade	14
10.15Figure A1: Timeline of airline bankruptcies in the United States, 1994 to 2006 . . .	15
10.16Figure A2: Total number of aircraft in bankruptcy	15
11 How to remember the paper	15

1 Big picture

Benmelech and Bergman ask whether a bankrupt firm can hurt otherwise solvent competitors by lowering the resale value of their collateral. The paper calls this mechanism the **collateral channel**. The idea is direct: when an airline enters bankruptcy, aircraft of the same model type become less valuable as collateral because the bankrupt airline may sell similar aircraft and because the bankrupt airline itself stops being a healthy potential buyer. Lower collateral value raises the credit spread on secured debt backed by those aircraft.

The empirical setting is U.S. airlines and aircraft-backed secured debt. This setting is unusually clean because airline secured debt tranches list the exact aircraft serving as collateral. The authors can therefore measure, for each debt tranche and date, whether the aircraft collateral is exposed to bankrupt airlines that operate the same model types.

The paper's central result is that bankruptcy-induced collateral shocks significantly raise the credit spreads of secured debt tranches issued by nonbankrupt airlines. The effect is strongest where theory says it should be strongest: for more junior tranches, higher loan-to-value tranches, weaker airlines, and less redeployable collateral. The results support a financial-accelerator interpretation in which bankruptcies lower industry collateral values, raise borrowing costs, and can amplify downturns.

2 Incremental contribution of the paper

1. **Moves beyond the bankrupt firm itself.** Prior bankruptcy research mainly studied firms that actually enter bankruptcy. This paper studies externalities imposed on other firms in the same industry.
2. **Identifies a specific contagion mechanism.** The paper distinguishes the collateral channel from generic negative industry news, product-market effects, lender effects, and investor fire-sale effects.
3. **Uses unusually detailed collateral-level data.** For each airline secured tranche, the authors identify the aircraft serial numbers and model types pledged as collateral.
4. **Exploits cross-sectional variation in collateral overlap.** The identification comes from whether a tranche is backed by aircraft types also operated by bankrupt airlines. This is more precise than simply comparing airlines before and after industry bankruptcies.
5. **Links bankruptcy to the cost of capital for solvent firms.** The main dependent variable is tranche credit spread, so the paper directly connects collateral impairment to financing costs.
6. **Shows why collateral redeployability matters in downturns.** Collateral is most valuable when many healthy potential buyers exist. When potential buyers are themselves distressed, collateral becomes less useful in supporting debt.
7. **Provides a financial-accelerator interpretation at the industry level.** Bankruptcies can depress collateral prices, reduce debt capacity, and potentially trigger further financial distress.

3 Research question and hypothesis

The main question is:

Do bankrupt firms impose negative externalities on nonbankrupt competitors by reducing the value of assets used as collateral?

The hypothesis has two related parts. First, bankruptcy increases the expected supply of used industry assets because bankrupt firms may liquidate or sell assets. Second, bankruptcy lowers demand for industry assets because the bankrupt firm is no longer a healthy buyer and may be financially unable to acquire assets. Both forces lower collateral values. If collateral supports debt capacity, then the credit spreads on secured debt backed by similar assets should increase.

In the airline setting, the mechanism is especially transparent. If a tranche is collateralized by Boeing 737 aircraft, then bankruptcies of airlines that operate Boeing 737s should be more relevant for that tranche than bankruptcies of airlines operating unrelated aircraft types. The paper therefore predicts that the spread effect should depend on *aircraft-model overlap*, not just on the number of bankrupt airlines in the industry.

4 Institutional setting: airline secured debt

The paper studies equipment trust certificates (ETCs), enhanced equipment trust certificates (EETCs), and pass-through certificates (PTCs). These instruments finance aircraft purchases and are secured by aircraft collateral. The most important features are:

- **Collateral is explicit.** Prospectuses identify the aircraft in the collateral pool, allowing the authors to connect each tranche to specific aircraft model types.
- **Tranches differ in seniority and LTV.** Senior tranches have more protection from collateral-value declines; junior tranches and high-LTV tranches should be more exposed.
- **Section 1110 protection strengthens creditor rights.** Aircraft-secured creditors have special rights in airline bankruptcy, making aircraft collateral central to pricing.
- **Some EETCs include liquidity facilities.** These facilities can pay interest for a limited period after default. The paper controls for them and checks that results are not driven by credit enhancers.

5 Data and construction of key variables

The sample begins with secured airline debt tranches issued in U.S. public markets. The authors match tranche information from SDC and prospectuses to aircraft-level data from Ascend CASE, transaction prices from Mergent FISD, issuer characteristics from Compustat, and airline bankruptcy dates from SDC Platinum restructuring data.

The final sample contains 18,327 transactions in 127 individual tranches issued by 12 airlines over 1994-2007. The dependent variable is the tranche credit spread: the yield-to-maturity on the tranche minus the duration-matched Treasury yield.

5.1 Bankrupt Buyers

For each aircraft model and date, the authors count the number of bankrupt airlines that operate that model. A tranche-level measure is then formed as a weighted average across aircraft in the collateral pool. This measure captures a **demand-side shock**: if potential buyers of the collateral are in bankruptcy, demand for that aircraft type is weaker.

5.2 Bankrupt Aircraft

For each aircraft model and date, the authors count how many aircraft of that model are operated by bankrupt airlines. A tranche-level weighted average is then constructed. This measure captures a **supply-side shock**: if bankrupt airlines operate many aircraft of the same model, the market may expect more similar aircraft to be sold.

5.3 Redeployability

Redeployability is measured by the number of healthy operators or healthy aircraft of the same model type. Highly redeployable aircraft have many possible users and should retain collateral value better. The paper uses this both as a control and as a way to test whether collateral shocks are weaker when assets are easier to redeploy.

6 Identification strategy

A simple comparison of spreads before and after airline bankruptcies would be hard to interpret because bankruptcies also reveal bad industry news. Benmelech and Bergman avoid this problem by using within-industry variation in collateral overlap. Their logic is:

1. Airline bankruptcies are industry-wide events and may contain common bad news.
2. But the collateral channel predicts a differential effect by aircraft model type.
3. A tranche backed by aircraft types used by bankrupt airlines should be hit harder than a tranche backed by aircraft types not used by bankrupt airlines.
4. This differential prediction helps separate collateral-specific price pressure from generic industry news.

The baseline regression relates tranche spread to the bankruptcy shock measures, collateral redeployability, issuer bankruptcy status, tranche controls, issuer controls, interest-rate controls, and fixed effects:

$$\text{Spread}_{i,a,t} = \beta_1 \log(1 + \text{Bankrupt Buyers}_{i,a,t}) + \beta_2 \text{Bankruptcy}_{i,a,t} + \beta_3 \log(1 + \text{Redeployability}_{i,a,t}) + \text{controls} + \varepsilon_{i,a,t}.$$

The preferred interpretation is that a positive β_1 means collateral exposed to bankrupt potential buyers commands a higher spread, even after controlling for issuer risk and industry conditions.

7 Main empirical findings

7.1 Bankruptcy shocks raise spreads

The univariate evidence already shows the pattern. Among nonbankrupt airlines, tranches with at least one bankrupt potential buyer have average spreads of 339 basis points, compared with 208 basis points for tranches with no bankrupt potential buyers. This is a large raw difference and is consistent with collateral impairment.

The regression evidence confirms the relationship. Across specifications, the *Bankrupt Buyers* and *Bankrupt Aircraft* measures are positively associated with tranche credit spreads. The effects are economically large, not just statistically significant.

7.2 Effects are stronger for exposed securities

The effect is larger for junior tranches and high-LTV tranches. This pattern is crucial because generic industry news should affect all tranches of an issuer more uniformly, while collateral-value risk should matter most for claims closer to the loss-absorbing margin.

The effect is also stronger for airlines in weaker financial health. If an airline is far from default, collateral liquidation value should matter less. If default probability is high, collateral value matters more. The results support this prediction.

7.3 Effects are temporary and tied to bankruptcy periods

The time-series tests show that the spread effect is concentrated while potential buyers are in bankruptcy. After firms exit bankruptcy, the effect largely fades. This is consistent with price pressure and temporary collateral-market impairment rather than a permanent reassessment of aircraft productivity.

7.4 Robustness checks reject alternative channels

The paper addresses several alternative explanations:

- **Industry conditions:** results hold after controlling for fuel prices, number of bankrupt airlines, bankrupt asset share, and healthy-airline measures.
- **Reverse causality:** results hold with aircraft-model-by-year controls and in subsamples restricted to common Airbus/Boeing aircraft.
- **Credit enhancement:** results are not driven by liquidity facilities.
- **Investor fire sales:** results hold for buy and sell transactions and for investment-grade and highly rated tranches.
- **Financial crisis:** excluding 2006-2007 does not eliminate the result.

8 Conceptual takeaway

The paper’s central contribution is not only that bankruptcies correlate with competitor financing costs. The deeper point is that the value of collateral is endogenous to the financial condition of other firms that use or might buy similar assets. When many industry participants are weak, collateral becomes less redeployable exactly when creditors most need it. This creates a mechanism through which downturns can amplify themselves.

9 Potential limitations and useful cautions

- **External validity.** Airlines are unusually suitable for the analysis because collateral is physical, standardized, and recorded precisely. The mechanism should generalize to other collateral-intensive industries, but the magnitude may differ.
- **Debt market sample.** The transactions are from Mergent FISD and involve insurance companies on at least one side. The paper addresses investor-channel concerns, but the data still capture a particular part of the bond market.
- **Collateral price is inferred.** The paper infers collateral-value impairment from tranche spreads rather than directly observing aircraft sale prices for every event.
- **Model overlap is a proxy.** Bankrupt Buyers and Bankrupt Aircraft are strong measures, but they are proxies for expected demand and supply pressure rather than literal realized purchases and sales.

10 Extracted figures and tables from the paper

10.1 Table I: EETC Structures

How to read. Table I gives examples of EETC issues by FedEx, Northwest Airlines, and Delta. It lists tranche size, yield, credit spread, maturity, credit ratings, LTV, and the collateral aircraft.

Interpretation. The table illustrates why this setting is powerful. Each tranche can be linked to a specific aircraft pool, and different tranches backed by the same pool can have different seniority and LTV. For example, FedEx 1998-1 has multiple tranches secured by the same MD-11F and A300 aircraft, but credit spreads rise as the tranche becomes less senior and LTV increases. This structure makes it possible to test whether collateral shocks matter more for junior and high-LTV claims.

10.2 Table II: Summary Statistics

How to read. Table II summarizes tranche characteristics, issuer characteristics, redeployability measures, and industry controls. Mean spread is 290 basis points, but the distribution is wide, with a maximum over 4,200 basis points. The mean LTV is 0.54. The average collateral pool has about 136 nonbankrupt potential operators and about 1,393 aircraft in the redeployability measure.

Interpretation. The summary statistics show that the sample contains substantial variation in credit risk, collateral redeployability, and industry distress. This variation is essential for the empirical strategy. The high dispersion in spreads means the data can capture economically meaningful changes in financing costs. The redeployability measures also show that some aircraft are much easier to sell or re-lease than others, which matters for secured credit pricing.

Table I
EETC Structures

This table displays the characteristics of three EETC issues by FedEx, Northwest Airlines, and Delta Airlines. Yield at issue, credit spread, credit ratings (both Moody's and S&P), and LTV are measured as the initial values at the date of the issue. Detailed variable definitions are provided in Appendix B.

EETC	Tranche	Issue Size	Yield at Issue (%)	Credit Spread (Basis Points)	Maturity	Moody's Rating	S&P Rating	LTV	Collateral
FedEx 1998-1	I-A	458.1	6.720	125	1/2022	Aa2	AAA	0.387	5 MD-11F 8 A300F4-605R
FedEx 1998-1	I-B	178.6	6.845	138	1/2019	A1	AA-	0.532	5 MD-11F 8 A300F4-605R
FedEx 1998-1	I-C	196.8	7.020	155	1/2016	Baa1	BBB+	0.688	5 MD-11F 8 A300F4-605R
NWA 1999-3	G	150.2	7.935	170	6/2019	Aaa	AAA	0.441	14 BAE Aero RJ85
NWA 1999-3	B	58.6	9.485	325	6/2015	Baa2	BBB	0.614	14 BAE Aero RJ85
NWA 1999-3	C	30.5	9.152	300	6/2010	Baa3	BBB-	0.691	14 BAE Aero RJ85
Delta 2002-1	G-1	586.9	6.718	153	1/2023	Aaa	AAA	0.519	17 B737-832 1 B757-232 8 B767-332ER 6 B767-432ER
Delta 2002-1	C	168.7	7.779	325	1/2012	Baa2	A-	0.611	17 B737-832 1 B757-232 8 B767-332ER 6 B767-432ER

Table I: EETC Structures

Table II
Summary Statistics

This table provides descriptive statistics for the variables used in the empirical analysis. Panel A displays tranche characteristics, Panel B provides airline characteristics, Panel C provides tranche redeployability characteristics, and Panel D presents industry-level controls. Variable definitions are provided in Appendix B.

	Mean	25th Percentile	Median	75th Percentile	Standard Deviation	Min	Max
Panel A: Tranche Characteristics							
Spread	290.2	153.6	229.4	330.8	311	16.9	4,206.6
Tranche Size (\$m)	274.4	127.0	207.1	385.8	181.2	3.5	828.8
Term to Maturity	16.9	14.5	18.1	20.2	4.5	1.7	24.3
Seniority	1.3	1.00	1.00	1.00	0.61	1.00	4.00
Call Provision	0.16	0.00	0.00	0.00	0.37	0.00	1.00
Amortized	0.68	0.00	1.00	1.00	0.47	0.00	1.00
Liquidity facility	0.75	0.00	1.00	1.00	0.43	0.00	1.00
LTV	0.54	0.41	0.49	0.66	0.16	0.33	0.89
Panel B: Airline Characteristics							
Size (\$m)	14,151.5	9,201.0	10,877.0	20,404.0	6,972.4	1,134.9	32,841.0
Market-to-Book	1.26	1.03	1.18	1.43	0.29	0.76	2.51
Profitability	8.24%	3.55%	10.39%	13.13%	6.76%	-12.10%	23.70%
Leverage	0.37	0.18	0.40	0.52	0.17	0.03	0.658
Panel C: Redeployability Measures							
Redeployability (# of aircraft)	1,392.9	424.7	1,046.2	2,345.4	1,016.0	72.0	4,264
Redeployability (# of operators)	135.9	48.8	87.0	223.5	99.9	7.0	431.0
Panel D: Airline Industry Variables							
Jet Fuel Price	107.8	70.9	84.4	146.6	55.1	29.6	280.5
Bankrupt Airlines	5.1	4.0	6.0	6.0	2.2	0.0	8.0
Healthy Airlines	62.0	58.0	62.0	65.0	4.1	51	73
Bankrupt Assets/ Total Assets	0.075	0.060	0.083	0.095	0.033	0.000	0.119
Healthy Assets/ Total Assets	0.925	0.905	0.917	0.940	0.033	0.881	1.000

Table II: Summary Statistics

10.3 Figure 1: Total number of bankrupt potential buyers for Boeing 737 and Boeing 747

How to read. Figure 1 plots, over time, how many bankrupt airlines operate Boeing 737 and Boeing 747 aircraft. These are potential buyers that become financially impaired while in bankruptcy.

Interpretation. The figure shows both common industry distress and aircraft-model-specific variation. Distress rises sharply after 2001, but the timing and intensity differ between the B737 and B747. This cross-model variation is the identification lever: if a secured tranche is backed by B737s, bankrupt B737 operators should matter more than bankrupt B747 operators.

10.4 Figure 2: Total number of Boeing 737 and Boeing 747 aircraft operated by bankrupt airlines in the United States

How to read. Figure 2 counts the number of B737 and B747 aircraft operated by bankrupt airlines. This captures the possible supply of similar aircraft that could enter the market through asset sales.

Interpretation. The B737 exposure is much larger than the B747 exposure during the major bankruptcy waves, reaching hundreds of aircraft. This supports the supply-side part of the collateral channel: some model types face much larger potential fire-sale pressure than others. A lender secured by aircraft that might be sold by bankrupt airlines should demand a higher spread.

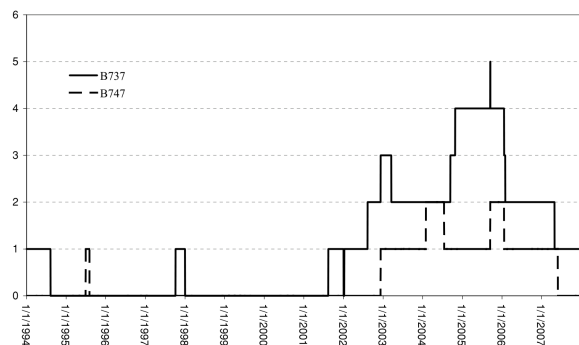


Figure 1. Total number of bankrupt potential buyers for Boeing 737 and Boeing 747. An airline is considered to be a potential buyer of a particular aircraft if in its fleet it operates aircraft of the same model type. Fleet data are obtained from the Ascend CASE database. Airline bankruptcy dates are obtained from SDC Platinum.

Figure 1: Total number of bankrupt potential buyers for Boeing 737 and Boeing 747

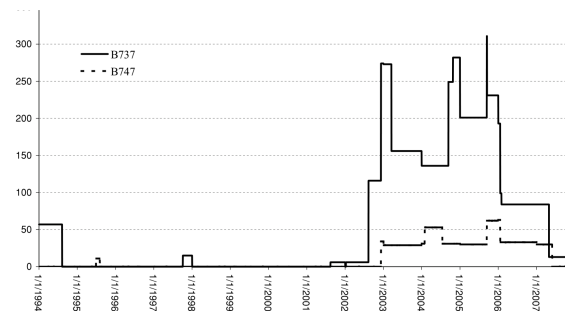


Figure 2. Total number of Boeing 737 and Boeing 747 aircraft operated by bankrupt airlines in the United States. Fleet data are obtained from the Ascend CASE database. Airline bankruptcy dates are obtained from SDC Platinum.

Figure 2: Total number of Boeing 737 and Boeing 747 aircraft operated by bankrupt airlines in the United States

10.5 Table III: Bankrupt Buyers and Bankrupt Aircraft Measures

How to read. Table III summarizes the two main bankruptcy-shock variables. Panel A gives the overall distribution, while Panels B and C show how the measures evolve over time.

Interpretation. The time pattern confirms that bankruptcy shocks are concentrated in the post-2001 period. The average Bankrupt Buyers measure rises from essentially zero before 2001 to over two in 2005, while Bankrupt Aircraft rises from almost zero to over 100 in 2005. This pattern lines up with airline distress after the 2001 downturn and the early-2000s bankruptcy wave. Importantly, the measures vary across tranche collateral pools, not just over calendar time.

10.6 Table IV: Bankruptcy, Bankrupt Buyers, and Tranche Credit Spreads: Univariate Analysis

How to read. Table IV gives simple mean comparisons. Panel A compares bankrupt and non-bankrupt issuers. Panel B compares nonbankrupt airlines with versus without bankrupt potential buyers. Panel C splits the comparison by tranche seniority.

Interpretation. Panel B is the simplest evidence for the collateral channel: among nonbankrupt airlines, tranches exposed to bankrupt potential buyers have mean spreads of 339 basis points versus 208 basis points for unexposed tranches. Panel C strengthens the interpretation: the spread difference grows as tranches become more junior. The most junior exposed tranches have a spread difference above 1,100 basis points. This pattern is hard to explain by generic industry bad news alone because seniority should not matter as much unless collateral recovery values are central.

Table III
Bankrupt Buyers and Bankrupt Aircraft Measures

This table provides descriptive statistics for the bankrupt buyers and bankrupt aircraft measures used in the empirical analysis. Panel A displays statistics for the entire sample, while Panels B and C provide statistics for different sample periods for each of the measures. Details on the construction of the *Bankrupt Buyers* and *Bankrupt Aircraft* measures are provided in Appendix B.

	Mean	25th Percentile	Median	75th Percentile	Standard Deviation	Min	Max
Panel A: Bankrupt Buyers and Number of Aircraft in Bankruptcy							
Bankrupt Buyers	0.809	0.0	0.269	1.571	1.027	0.0	5.0
Bankrupt Aircraft	43.860	0.0	2.628	86.177	63.275	0.0	311.0
Panel B: Bankrupt Buyers over Time							
1994–2000	0.013	0.0	0.0	0.0	0.077	0.0	1.0
2001	0.324	0.0	0.244	0.528	0.348	0.0	1.0
2002	0.578	0.0	0.396	1.0	0.648	0.0	3.0
2003	1.411	0.608	1.725	2.161	0.872	0.0	3.0
2004	1.604	0.533	1.732	2.299	1.171	0.0	4.0
2005	2.058	0.105	2.167	3.336	1.483	0.0	5.0
2006	1.023	0.0	1.309	1.732	0.864	0.0	4.0
2007	0.372	0.0	0.0	0.619	0.514	0.0	2.0
Panel C: Bankrupt Aircraft over Time							
1994–2000	0.227	0.0	0.0	0.0	1.859	0.0	57
2001	3.042	0.0	1.461	5.751	3.817	0.0	17
2002	23.832	0.0	2.388	52.851	39.208	0.0	274
2003	93.286	31.529	111.585	128.242	61.598	0.0	273
2004	91.445	13.0	96.509	118.132	70.884	0.0	282
2005	110.021	3.206	117.976	182.282	83.939	0.0	311
2006	63.323	0.0	73.0	93.846	54.236	0.0	264
2007	14.885	0.0	0.0	8.947	34.126	0.0	181

Table III: Bankrupt Buyers and Bankrupt Aircraft Measures

Table IV
Bankruptcy, Bankrupt Buyers, and Tranche Credit Spreads: Univariate Analysis

This table provides univariate analysis of tranche credit spreads: Panel A segments credit spreads of tranches of nonbankrupt and bankrupt airlines, Panel B focuses on nonbankrupt airlines and compares tranche credit spreads of tranches with a positive bankrupt buyer measure and those with a bankrupt buyer measure equal to zero, while Panel C stratifies the analysis in Panel B by tranche seniority, and reports means and *t*-statistics for *t*-tests of equal means using standard errors that are clustered at the tranche level.

Panel A: Tranche Credit Spreads of Bankrupt and Nonbankrupt Airlines: Summary Statistics							
	Mean	10th Percentile	25th Percentile	Median	75th Percentile	Standard Deviation	Observations
Bankrupt Airlines	531.7	135.1	188.4	317.4	592.4	649.9	1,011
Nonbankrupt Airlines	276.1	94.2	152.6	226.1	322.7	273.3	17,316
Difference	255.6						
<i>T</i> -test for equal means	(2.81)						
Panel B: Tranche Credit Spreads of Nonbankrupt Airlines: Summary Statistics							
	Mean	10th Percentile	25th Percentile	Median	75th Percentile	Standard Deviation	Observations
Bankrupt Buyers > 0	339.0	135.9	197.7	271.7	363.4	336.1	8,992
No Bankrupt Buyers	208.0	75.4	129.4	177.1	253.3	156.3	8,324
Difference	131.0						
<i>T</i> -test for equal means	(7.48)						
Panel C: Tranche Credit Spreads of Nonbankrupt Airlines and Seniority: Means and <i>T</i> -tests							
	1	2	3	4	Diff (2-1) (<i>T</i> -test)	Diff (3-1) (<i>T</i> -test)	Diff (4-1) (<i>T</i> -test)
Bankrupt Buyers > 0	302.5	419.2	474.9	1,444.9	116.7	172.47	1,142.5
(Observations)	(6,755)	(1,613)	(590)	(34)	(3.79)	(2.81)	(7.54)
No Bankrupt Buyers	207.5	223.7	177.5	332.0	16.3	30.03	124.5
(Observations)	(6,481)	(1,187)	(625)	(31)	(0.51)	(1.86)	(4.77)
Difference	95.0	195.5	297.4	1,112.9			
<i>T</i> -test for equal means	(5.70)	(4.08)	(5.15)	(6.79)			

Table IV: Bankruptcy, Bankrupt Buyers, and Tranche Credit Spreads: Univariate Analysis

10.7 Table V: Bankruptcy and Collateral

How to read. Table V estimates credit spread regressions using the two collateral-shock measures. Panel A uses Bankrupt Buyers, Panel B uses Bankrupt Aircraft, and Panel C translates the estimates into basis-point magnitudes.

Interpretation. Both measures are positive and significant across specifications. The result survives year fixed effects, airline fixed effects, tranche fixed effects, and interactions with the issuer's own bankruptcy status. Panel C shows the economic magnitude: moving from the 25th to the 75th percentile of Bankrupt Buyers raises spreads by roughly 147 to 238 basis points in the fixed-effect specifications. This is a large increase in the cost of secured debt financing. The

negative coefficients on redeployability in many columns are also consistent with the same theory: collateral that can be used by more healthy operators supports lower spreads.

10.8 Table VI: Bankruptcy and Collateral: Credit Spreads of Nonbankrupt Airlines

How to read. Table VI restricts the sample to tranches of nonbankrupt airlines. This directly tests whether bankruptcies affect solvent competitors, not just the bankrupt issuer's own debt.

Interpretation. The positive Bankrupt Buyers coefficients remain statistically and economically meaningful after excluding bankrupt issuers. This is one of the paper's core tables. It shows that the collateral channel is an externality: the financing costs of solvent airlines rise when bankrupt airlines overlap with their collateral pool. The control variables behave sensibly: higher profitability lowers spreads, higher leverage raises spreads, liquidity facilities lower spreads, and seniority lowers risk.

Table V Bankruptcy and Collateral						
The table presents coefficient estimates and standard errors (in parentheses) for credit spread regressions. Panel A uses the <i>Bankrupt Buyers</i> measure while Panel B uses the <i>Bankrupt Aircraft</i> measure. For each specification, Panel C of the table provides estimates of the magnitude of the economic effect of either a one standard deviation move or a 25th to 75th percentile movement in the <i>Bankrupt Buyers</i> and <i>Bankrupt Aircraft</i> measures on tranche credit spread. All regressions include an intercept, yield curve, and default spread controls (short rate, term spread, and default spread). Standard errors are calculated by clustering at the tranche level. Variable definitions are provided in Appendix B. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.						
Panel A: Bankrupt Buyers						
Dependent Variable =	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread
Bankrupt Buyers	246.491*** (28.700)	151.619*** (31.387)	106.162*** (27.299)	93.614*** (27.124)	119.030*** (28.302)	118.937*** (27.505)
Bankruptcy	126.600** (76.610)	168.670** (67.129)	144.448** (65.772)	21.485 (30.607)	184.451*** (70.037)	565.409*** (28.287)
Redeployability (operators)	-71.718*** (14.019)	-50.960*** (13.592)	-97.386*** (27.094)	-80.352** (26.466)	-5.850 (58.415)	20.889 (56.732)
Adjusted R ²	0.16	0.22	0.27	0.28	0.38	0.48
Panel B: Bankrupt Aircraft						
Bankrupt Aircraft	53.790*** (10.796)	26.479*** (7.765)	20.706*** (6.117)	20.323*** (6.124)	25.522*** (5.994)	26.574*** (5.683)
Bankruptcy	116.066 (80.416)	172.344** (68.805)	133.336* (66.294)	9.531 (29.499)	182.755** (70.569)	566.077*** (26.773)
Redeployability (aircraft)	-46.139*** (10.796)	-21.955** (11.323)	-83.065*** (21.943)	-73.395*** (21.183)	-60.464 (62.818)	-27.262 (61.984)
Adjusted R ²	0.15	0.21	0.27	0.28	0.37	0.48
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year	No	No	No	No	No	No
Airline	No	No	No	No	No	No
Airline × Bankruptcy	No	No	No	No	No	No
Tranche × Bankruptcy	No	No	No	No	No	No
# of Tranches	127	127	127	127	127	127
# of Airlines	12	12	12	12	12	12
Observations	18,327	18,327	18,327	18,327	18,327	18,327
Panel C: Magnitude of the Collateral Channel (in Basis Points)						
Bankrupt Buyers						
One σ change	253.15	155.71	109.03	96.14	122.24	122.15
25%-75% change	387.24	238.19	166.78	147.07	187.00	186.85
Bankrupt Aircraft						
One σ change	114.45	56.34	44.06	43.24	54.30	56.54
25%-75% change	240.33	118.31	92.51	90.80	114.03	118.73

Table V: Bankruptcy and Collateral

Table VI Bankruptcy and Collateral: Credit Spreads of Nonbankrupt Airlines						
The table presents coefficient estimates and standard errors (in parentheses) for credit spread regressions. All regressions include an intercept, yield curve, and default spread controls (short rate, term spread, and default spread) and year fixed effects. Standard errors are calculated by clustering at the tranche level. Variable definitions are provided in Appendix B. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.						
Dependent Variable =	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread
Bankrupt Buyers	67.340*** (29.409)	56.389** (28.274)	64.666** (27.984)			
Redeployability (operators)	-36.463** (16.436)	-79.354*** (22.800)	-32.604 (79.046)			
Bankrupt Aircraft				9.230 (7.035)	11.681* (6.645)	13.678** (5.686)
Redeployability (aircraft)				-22.541* (13.980)	-68.859*** (17.178)	-20.251 (82.924)
Fuel Price	30.945 (36.120)	30.922 (37.415)	18.396 (32.972)	42.151 (36.609)	35.107 (37.588)	30.562 (33.405)
Number Bankrupt	12.076** (5.857)	13.975** (5.908)	12.034** (5.565)	15.519*** (5.713)	16.383*** (5.722)	14.042** (5.476)
Amortizing	-146.340*** (29.991)	-149.022*** (29.355)		-145.937*** (30.153)	-148.185*** (29.385)	
Liquidity Facility	-123.517*** (39.479)	-115.703*** (41.827)		-123.579*** (39.518)	-112.787*** (39.518)	
Seniority	57.867** (24.171)	75.399*** (25.042)		55.261** (23.917)	74.632** (24.579)	
Tranche Size	-52.249*** (17.691)	-36.730* (20.679)		-54.385*** (17.202)	-38.217* (19.989)	
Call Provision	8.966 (26.172)	13.016 (25.728)		10.611 (26.129)	13.270 (25.763)	
Term-to-Maturity	7.815*** (2.753)	9.761*** (3.034)		7.648*** (2.757)	9.637*** (3.024)	
Airline Size	39.289 (29.077)	-18.290 (54.104)	29.882 (69.825)	41.329 (28.935)	-5.994 (55.018)	15.310 (72.974)
Market-to-Book	107.908** (41.224)	170.989*** (37.813)	180.563*** (37.770)	102.698** (41.353)	155.842*** (37.859)	181.503*** (37.465)
Profitability	-1,003.526*** (192.222)	-886.778*** (201.308)	-967.063*** (242.314)	-1,073.334*** (186.180)	-848.450*** (205.402)	-1,008.489*** (240.932)
Leverage	400.752*** (97.933)	470.405*** (136.970)	521.011*** (132.325)	405.773*** (95.483)	472.754*** (135.697)	545.577*** (132.010)
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year	No	No	No	No	No	No
Airline	No	No	No	No	No	No
Tranche	No	No	Yes	No	No	Yes
# of Tranches	126	126	126	126	126	126
# of Airlines	12	12	12	12	12	12
Adjusted R ²	0.28	0.30	0.41	0.28	0.30	0.41
Observations	16,877	16,877	16,877	16,877	16,877	16,877

Table VI: Bankruptcy and Collateral: Credit Spreads of Nonbankrupt Airlines

10.9 Table VII: The Time Series of the Collateral Channel

How to read. Table VII adds variables measuring recent exits from bankruptcy. The goal is to see whether the effect continues after bankrupt airlines exit Chapter 11.

Interpretation. The current bankruptcy measures remain positive, but the post-bankruptcy variables are generally not positive and often become negative in the 120-240 day window. This

suggests that the spread increase is tied to the period in which potential buyers are financially constrained or aircraft sale pressure is active. The pattern supports a temporary price-pressure or collateral-value channel rather than a permanent aircraft-model-quality shock.

10.10 Table VIII: The Collateral Channel and Industry Conditions

How to read. Table VIII adds industry controls such as fuel prices, number of bankrupt airlines, the bankrupt-airline asset share, the number of healthy airlines, and healthy-airline asset shares.

Interpretation. The collateral-shock coefficients remain positive after controlling for broad industry conditions. This is important because the main threat to identification is that bankruptcies reveal bad industry news. The table shows that the results are not merely picking up the level of industry distress. The effect depends on whether the tranche's collateral overlaps with the bankrupt firms' fleets, conditional on general industry deterioration.

Table VII
The Time Series of the Collateral Channel

The table presents coefficient estimates and standard errors (in parentheses) for credit spread regressions. All regressions include an intercept, yield curve, and default spread controls (short rate, term spread, and default spread), as well as tranche, airline, and industry controls and year fixed effects. Standard errors are calculated by clustering at the tranche level. Variable definitions are provided in Appendix B. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable =	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread
Redeployability (operators)	-34.860** (16.041)	-74.755*** (22.024)	29.440 (79.885)			
Bankrupt Buyers	76.343*** (28.470)	65.867** (28.015)	66.696** (28.272)			
Ex-Bankrupt Buyers _{0,120}	26.146 (22.402)	13.852 (22.237)	25.975 (19.782)			
Ex-Bankrupt Buyers _{120,240}	-70.954*** (17.302)	-75.615*** (17.644)	-72.041*** (14.136)			
Ex-Bankrupt Buyers _{240,360}	-29.938 (26.783)	-34.875 (27.989)	-39.442* (28.806)			
Redeployability (aircraft)				-20.764 (13.839)	-68.985*** (16.551)	-11.066 (80.613)
Bankrupt Aircraft				13.108** (6.738)	13.167** (6.384)	15.525*** (5.738)
Ex-Bankrupt Aircraft _{0,120}				0.157 (0.398)	0.187 (0.401)	0.329 (0.347)
Ex-Bankrupt Aircraft _{120,240}				-1.177*** (0.273)	-1.098*** (0.275)	-1.063*** (0.204)
Ex-Bankrupt Aircraft _{240,360}				-0.743* (0.402)	-0.673 (0.431)	-0.782** (0.367)
Fixed Effects						
Year	Yes	Yes	Yes	Yes	Yes	Yes
Airline	No	Yes	No	No	Yes	No
Tranche	No	No	Yes	No	No	Yes
# of Tranches	126	126	126	126	126	126
# of Airlines	12	12	12	12	12	12
Adjusted R ²	0.28	0.31	0.42	0.28	0.30	0.42
Observations	16,877	16,877	16,877	16,877	16,877	16,877

Table VII: The Time Series of the Collateral Channel

Table VIII
The Collateral Channel and Industry Conditions

The table presents coefficient estimates and standard errors (in parentheses) for credit spread regressions. All regressions include an intercept, yield curve, and default spread controls (short rate, term spread, and default spread), and tranche and year fixed effects. Tranche controls are not included in the explanatory variables as they are absorbed by the tranche fixed effects. Standard errors are calculated by clustering at the tranche level. Variable definitions are provided in Appendix B. *** and ** denote statistical significance at the 1% and 5% levels, respectively.

Dependent Variable =	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread
Bankrupt Buyers	64.666** (27.984)	64.656** (27.920)	73.134** (29.550)			
Redeployability (operators)	32.604 (79.047)	32.269 (79.015)	25.878 (80.208)			
Bankrupt Aircraft				13.678** (5.868)	13.676** (5.679)	14.972** (6.023)
Redeployability (aircraft)				-20.251 (82.924)	-20.722 (82.891)	-22.148 (84.435)
Fuel Price	18.396 (32.972)	18.005 (32.900)	17.955 (32.040)	30.562 (33.407)	30.103 (33.336)	31.082 (32.553)
Number Bankrupt	12.034** (5.565)					
Number Bankrupt/Total		831.199** (376.357)				
Bankrupt Assets/Total Assets			7.404 (110.016)			
Number Healthy				-14.042** (5.476)		
Healthy/Total					-969.258** (371.263)	
Healthy Assets/Total Assets						-7.937 (100.998)
Airline Size	29.882 (69.824)	30.117 (69.745)	39.117 (69.717)	5.310 (72.974)	5.580 (72.910)	15.343 (72.074)
Market-to-Book	180.563*** (37.770)	180.651*** (37.761)	176.786** (37.573)	181.503*** (37.465)	181.561*** (37.453)	175.090*** (37.233)
Profitability	-964.063*** (242.314)	-965.595*** (242.353)	-942.286*** (241.644)	-1,008.489*** (240.932)	-1,010.179*** (240.968)	-981.395*** (239.619)
Leverage	521.011*** (132.326)	520.949*** (132.214)	503.475 (133.508)	545.577*** (132.010)	545.414*** (131.928)	529.795*** (133.248)
Fixed Effects	Tranche+ Year	Tranche+ Year	Tranche+ Year	Tranche+ Year	Tranche+ Year	Tranche+ Year
# of Tranches	126	126	126	126	126	126
# of Airlines	12	12	12	12	12	12
Adjusted R ²	0.42	0.41	0.41	0.41	0.41	0.41
Observations	16,877	16,877	16,877	16,877	16,877	16,877

Table VIII: The Collateral Channel and Industry Conditions

10.11 Table IX: Bankruptcy, Collateral, Tranche Seniority, and LTV

How to read. Table IX tests whether the collateral channel is stronger for junior tranches and higher-LTV tranches. It uses interactions with seniority and LTV and also splits the sample.

Interpretation. The interaction results match the theory. Junior claims and high-LTV claims are more sensitive to bankrupt potential buyers because they are closer to the loss margin if collateral values fall. Senior claims have a larger cushion and are less exposed. This table is especially persuasive because it identifies where the mechanism should be strongest, rather than

simply showing an average spread increase.

10.12 Table X: The Collateral Channel and Airlines' Financial Strength

How to read. Table X asks whether collateral shocks matter more when the issuing airline is financially weak. It uses interactions with profitability and predicted bankruptcy probability, then splits by seniority and LTV.

Interpretation. The negative interaction between Bankrupt Buyers and profitability for junior/high-LTV tranches means that the collateral channel becomes stronger when issuer profitability is lower. This is exactly what secured credit theory predicts. If default is unlikely, collateral recovery value is less central; if default risk rises, lenders care much more about whether the aircraft can be sold to healthy buyers. The predicted-bankruptcy interactions provide a more direct version of the same idea.

Table IX
Bankruptcy, Collateral, Tranche Seniority, and LTV

The table presents coefficient estimates and standard errors (in parentheses) for credit spread regressions. Regressions are estimated separately for senior and junior tranches (columns 3 and 4), and for tranches with LTV lower than 0.5 (column 7) and higher than 0.5 (column 8). All regressions include an intercept, yield curve, and default spread controls (short rate, term spread, and default spread) and year fixed effects. Columns 1 and 5 include airline fixed effects, and columns 2, 3, 6, 7, and 8 include tranche fixed effects. Standard errors are calculated by clustering at the tranche level. Variable definitions are provided in Appendix B. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable =	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread
			Senior	Junior				
							LTV	
							<0.5	≥0.5
Bankrupt	-163.798***	-139.771***	41.566*	429.922**	-334.580***	-271.665**	49.282***	116.535**
Buyers	(51.636)	(53.375)	(24.483)	(164.483)	(103.743)	(110.191)	(16.484)	(48.112)
>Seniority	171.992***	154.430***						
	(37.852)	(41.392)						
Redeployability (operators)	-79.995***	35.396	44.476	-330.419	-81.905***	23.951	80.757***	66.882
	(22.011)	(77.082)	(82.536)	(210.222)	(23.534)	(82.602)	(27.06)	(123.474)
Bankrupt Buyers >LTV					729.163***	628.430***		
					(228.682)	(240.669)		
Fuel Price	34.189	19.815	24.803	-140.540	35.070	28.017	32.596	-11.179
	(36.318)	(32.921)	(33.465)	(179.532)	(86.559)	(31.504)	(24.357)	(58.422)
Industry	11.646**	10.359*	11.502*	30.160	14.282***	11.658**	0.226	16.025**
	(5.839)	(5.531)	(5.806)	(32.122)	(5.360)	(5.206)	(3.677)	(8.072)
Bankruptcy Amortizing	-135.727***				-140.008***			
	(26.262)				(26.464)			
Liquidity facility	-103.482***				-117.690			
	(30.017)				(78.778)			
Seniority	14.127				55.155**			
	(14.291)				(24.483)			
LTV					-175.823			
					(198.648)			
Tranche Size	-24.237				-28.370			
	(17.534)				(19.199)			
Call Provision	12.040				6.223			
	(22.920)				(23.968)			
Term-to-Maturity	10.729***				5.887*			
	(2.968)				(3.049)			
Airline Size	22.731	55.280	11.546	488.523*	-80.246	-12.202	191.157***	-67.034
	(52.601)	(66.318)	(69.019)	(277.255)	(57.500)	(75.626)	(44.730)	(119.146)
Market-to-Book	171.933***	179.921***	160.28***	1,116.765*	181.280**	189.618**	46.052	237.571***
	(36.531)	(35.866)	(34.732)	(618.322)	(41.091)	(38.043)	(43.905)	(47.302)
Profitability	-1,011.954***	-1,066.630***	-916.369***	-1,404.840	-800.409**	-801.149***	-614.626**	-1,822.178***
	(184.776)	(220.806)	(230.722)	(996.173)	(183.056)	(226.702)	(189.368)	(473.238)
Leverage	417.267***	474.445***	474.647***	2,183.932	477.450***	500.080**	32.809	592.249*
	(128.517)	(127.056)	(130.849)	(1,491.463)	(137.739)	(111.453)	(113.296)	(293.031)
Fixed Effects	Airline+ Year	Tranche+ Year	Tranche+ Year	Tranche+ Year	Airline+ Year	Tranche+ Year	Tranche+ Year	Tranche+ Year
# of Tranches	126	126	106	20	116	116	38	78
# of Airlines	12	12	9	5	9	9	7	9
Adjusted R ²	0.33	0.43	0.41	0.49	0.34	0.45	0.57	0.45
Observations	16,877	16,877	15,649	1,228	16,174	16,174	8,285	7,889

Table IX: Bankruptcy, Collateral, Tranche Seniority, and LTV

10.13 Table XI: The Collateral Channel and Tranche Fleet

How to read. Table XI addresses reverse causality and aircraft-specific productivity shocks. It adds aircraft-model-by-year fixed effects, restricts collateral to Airbus/Boeing or Boeing-only fleets, and controls for wide-body aircraft interacted with month-year fixed effects.

Interpretation. The concern is that a particular aircraft type might become less productive, causing both its operators to enter bankruptcy and its collateral value to fall. Table XI makes that interpretation less plausible. The results remain for common general-purpose aircraft and after

Table X
The Collateral Channel and Airlines' Financial Strength

The table presents coefficient estimates and standard errors (in parentheses) for credit spread regressions. Regressions are estimated separately based on seniority (columns 1 vs. 2, and columns 5 vs. 6), and LTV (columns 3 vs. 4, and columns 7 vs. 8). All regressions include an intercept, yield curve, and default spread controls (short rate, term spread, and default spread), as well as tranche and year fixed effects. Standard errors are calculated by clustering at the tranche level. Variable definitions are provided in Appendix B. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable =	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread
			Seniority Level	LTV	Seniority Level	LTV	Seniority Level	LTV
			Senior	Junior	Senior	Junior	Senior	Junior
Bankrupt	84.618**	873.766***	49.684*	312.007***	62.537	450.407**	93.076***	45.452**
Buyers	(40.390)	(240.505)	(25.142)	(86.494)	(39.957)	(168.984)	(21.073)	(80.668)
Redeployability (operators)	53.370	-355.682**	80.796**	86.712	52.873	-172.991	48.316	78.510
	(81.042)	(169.908)	(26.496)	(115.963)	(66.243)	(145.143)	(31.294)	(108.399)
Bankrupt Buyers >Profitability	-620.859*	-6,782.759***	-8.591	-2,159.232***				
	(341.277)	(2,005.524)	(36.351)	(577.889)				
Bankrupt Buyers >Pr(Bankruptcy)					57.179	-71.805	-147.238**	472.031*
					(137.833)	(662.581)	(63.233)	(280.010)
Fuel Price	19.618	-172.159	32.543	-13.641	-18.047	-196.045	14.168	-56.953
	(33.372)	(186.495)	(23.992)	(56.464)	(29.475)	(133.623)	(24.511)	(51.225)
Industry	12.384**	42.083	0.266	16.940**	-0.548	17.797	-3.927	8.017
	(5.807)	(32.475)	(4.223)	(8.117)	(5.547)	(21.531)	(4.974)	(9.102)
Bankruptcy Pr(Bankruptcy)					30.460	498.291	53.253	97.183
					(54.969)	(476.970)	(36.658)	(139.682)
Airline Size	50.254	705.267***	191.194***	123.501				
	(76.971)	(247.669)	(44.997)	(128.668)				
Market-to-Book	144.177**	1,109.565**	46.201	184.908***				
	(34.441)	(497.782)	(46.534)	(40.491)				
Profitability	-754.646***	571.176	-611.588***	-1,428.775***				
	(218.162)	(1,660.159)	(169.520)	(420.363)				
Leverage	426.302**	1,862.589	33.434	187.569				
	(134.184)	(1,537.861)	(122.091)	(301.548)				
Fixed Effects	Tranche+ Year	Tranche+ Year	Tranche+ Year	Tranche+ Year	Tranche+ Year	Tranche+ Year	Tranche+ Year	Tranche+ Year
# of Tranches	106	20	38	78	106	20	38	78
# of Airlines	9	5	7	9	9	5	7	9
Adjusted R ²	0.41	0.51	0.57	0.46	0.38	0.46	0.55	0.42
Observations	15,649	1,228	8,285	7,889	15,649	1,228	8,285	7,889

Table X: The Collateral Channel and Airlines' Financial Strength

absorbing model-specific time variation. The positive Bankrupt Buyers and Bankrupt Aircraft coefficients therefore appear to reflect bankruptcy-induced collateral-market pressure rather than news about a specific aircraft model.

10.14 Table XII: Bankruptcy Collateral and Investment Grade

How to read. Table XII separates buy and sell transactions and examines investment-grade, Aaa-Aa3, and Aaa tranches. The last columns exclude 2006-2007 to address concerns about the financial crisis period.

Interpretation. The Bankrupt Buyers coefficient remains positive for both buy and sell transactions and for highly rated tranches. This weakens the explanation that the results are driven by insurance-company fire sales of downgraded or risky securities. The results also hold when the authors exclude the years closest to the 2007-2009 crisis. The table therefore supports the view that the spread effect is tied to collateral values rather than investor regulation, rating thresholds, or crisis-specific forced selling.

Table XI
The Collateral Channel and Tranche Fleet

The table presents coefficient estimates and standard errors (in parentheses) for credit spread regressions. All regressions include an intercept, yield curve, and default spread controls (short rate, term spread, and default spread), and tranche fixed effects. Columns 1 and 2 also include aircraft model $\times$ year fixed effects. Columns 3 and 4 focus on aircraft manufactured by either Airbus and Boeing (column 3) or Boeing only (column 4). The last two columns of the table also include a dummy variable for wide-body aircraft interacted with month-year fixed effects. Tranche controls are not included in the explanatory variables as they are absorbed by the tranche fixed effects. Standard errors are calculated by clustering at the tranche level. Variable definitions are provided in Appendix B. *** and ** denote statistical significance at the 1% and 5% levels, respectively.

Dependent Variable=	Tranche Spread	Tranche Spread	<i>Airbus and Boeing</i>		Tranche Spread	Tranche Spread
			Spread	Spread		
Bankrupt Buyers	164.072*** (40.917)		103.499*** (36.759)	112.696*** (41.124)	123.376*** (42.468)	
Redeployability (operators)	15.783 (103.268)		-38.579 (81.353)	77.172 (97.071)	17.332 (69.988)	
Bankrupt Aircraft		30.028*** (8.772)			26.121*** (8.764)	
Redeployability (aircraft)		41.212 (101.099)			-27.966 (70.416)	
Fuel Price	15.128 (35.081)	30.548 (34.520)	61.196 (40.156)	-18.773 (39.344)	35.010 (71.103)	49.094 (70.261)
Number Bankrupt	-2.907 (5.681)	3.506 (5.384)	7.948 (7.833)	0.946 (7.012)	-22.516 (13.744)	-16.864 (12.319)
Airline Size	83.260 (179.591)	30.844 (179.205)	122.048 (94.711)	-28.924 (116.073)	-80.478 (96.941)	-86.821 (102.811)
Market-to-Book	47.254 (164.964)	14.459 (164.320)	146.193*** (46.055)	217.955** (91.490)	106.684** (43.154)	113.051** (43.827)
Profitability	-1.369.364*** (619.761)	-1.521.139** (642.168)	-2.149.84*** (417.980)	-1.846.818** (703.254)	-1.259.009*** (303.186)	-1.285.796*** (303.611)
Leverage	252.650 (467.657)	268.310 (477.368)	-187.420 (185.463)	-214.891 (418.264)	478.857*** (145.710)	514.745*** (140.212)
Fixed Effects	Tranche+ Year $\times$ Aircraft Model	Tranche+ Year $\times$ Aircraft Model	Tranche+ Year	Tranche+ Year	Tranche+ Month- Year $\times$ Widebody	Tranche+ Month- Year $\times$ Widebody
# of Tranches	126	126	105	69	126	126
# of Airlines	12	12	8	7	12	12
Adjusted R ²	0.47	0.47	0.40	0.43	0.51	0.51
Observations	16,877	16,877	14,204	9,368	16,877	16,877

Table XI: The Collateral Channel and Tranche Fleet

Table XII
Bankruptcy Collateral and Investment Grade

The table presents coefficient estimates and standard errors (in parentheses) for credit spread regressions. Regressions are estimated separately for buy vs. sell transactions. All regressions include an intercept, yield curve, and default spread controls (short rate, term spread, and default spread), as well as airline and year fixed effects. The first two columns compare buy vs. sell transactions of tranches with Moody's investment grade credit ratings. Columns 3 and 4 compare buy vs. sell transactions of tranches with Moody's ratings that are between Aaa and Aa3. Columns 5 and 6 compare buy vs. sell transactions of tranches with Aaa Moody's credit rating. The last two columns compare buy vs. sell transactions of tranches with Moody's investment grade credit ratings for the 1994 to 2005 period, excluding the years 2006 and 2007. Standard errors are calculated by clustering at the tranche level. Variable definitions are provided in Appendix B. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable =	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread	Tranche Spread
Transaction Type: Rating:	Buy Investment Grade	Sell Investment Grade	Buy Aaa-Aa3	Sell Aaa-Aa3	Buy Aaa	Sell Aaa	Buy Investment Grade	Sell Investment Grade
							Sample Period	
							1994-2005	1994-2005
Bankrupt	48.801*** (16.680)	28.436*** (16.831)	43.177*** (11.165)	12.188*** (3.379)	31.962*** (7.533)	39.301*** (12.343)	64.910*** (13.259)	32.211*** (10.393)
Buyers	-0.222** (0.111)	-0.442*** (0.156)	-0.578** (0.284)	-0.278*** (0.093)	-0.507*** (0.107)	-0.482*** (0.194)	-0.269** (0.107)	-0.545*** (0.146)
Redeployability (operators)	36.150 (34.020)	6.791 (31.589)	156.707 (96.294)	13.314 (21.353)	-86.534* (47.505)	41.391 (60.800)	6.416 (21.428)	4.472 (37.453)
Fuel Price	3.029 (5.262)	-5.618 (5.264)	-25.724 (18.225)	3.739 (4.279)	-9.492 (7.247)	-19.736 (17.619)	6.262 (4.932)	-1.925 (5.575)
Industry Bankruptcy	-60.338*** (21.925)	-117.907*** (31.827)	-334.867*** (148.341)	225.178 (134.912)			-71.500*** (18.818)	-121.490*** (31.718)
Amortizing	-35.768* (18.637)	-28.082** (13.238)					-49.237*** (17.273)	-46.239* (23.818)
Liquidity	28.724** (11.972)	18.541*** (13.576)	72.287 (91.376)	-24.276 (45.892)			33.440*** (12.108)	27.793* (14.565)
Seniority	-19.156 (12.685)	-26.360* (15.353)	43.698* (24.614)	-79.007** (32.571)			-15.443 (12.787)	-19.493 (15.161)
Tranche Size	-46.212* (26.038)	-30.865 (28.396)	-24.608*** (4.359)	-4.278 (5.010)			-42.646 (25.834)	-30.022 (27.365)
Call Provision	2.630 (2.599)	7.817*** (2.568)	32.431** (14.141)	-19.504* (11.111)			4.395** (2.034)	7.733*** (2.663)
Term-to-Maturity	-34.097 (67.988)	75.192 (48.534)	-458.837 (329.577)	106.547 (69.108)	-292.743 (251.141)	-516.561*** (97.509)	15.698 (50.102)	122.178** (48.699)
Airline Size	13.758 (32.354)	89.397*** (29.457)	-146.136 (139.575)	85.104*** (18.108)	-124.277 (135.766)	-181.185** (78.374)	14.335 (36.162)	50.870** (22.563)
Market-to-Book	-436.356*** (157.622)	163.745 (214.282)	809.942 (1,010.792)	99.093 (209.242)	-1,776.354*** (241.397)	-1,191.614*** (58.610)	-480.859*** (143.742)	58.557 (210.623)
Profitability	158.462 (166.102)	131.025 (85.686)	504.531 (430.307)	1.292 (106.396)	612.351*** (35.894)	58.698** (22.308)	80.305 (151.204)	98.440 (86.284)
Leverage								
Fixed Effects	Airline+ Year	Airline+ Year	Airline+ Year	Airline+ Year	Airline+ Year	Airline+ Year	Airline+ Year	Airline+ Year
# of Tranches	93	96	17	17	5	4	93	95
# of Airlines	8	9	7	6	4	3	8	9
Adjusted R ²	0.34	0.33	0.64	0.86	0.96	0.79	0.38	0.32
Observations	2,374	7,196	591	966	74	259	2,285	6,101

Table XII: Bankruptcy Collateral and Investment Grade

10.15 Figure A1: Timeline of airline bankruptcies in the United States, 1994 to 2006

How to read. Figure A1 plots the timing and duration of U.S. airline bankruptcies. Each horizontal segment marks a bankruptcy spell for an airline.

Interpretation. The figure shows that airline bankruptcy episodes are long and clustered. The early 2000s contain several overlapping bankruptcy spells, including major airlines. This clustering explains why the collateral channel can matter at the industry level: when multiple firms are simultaneously distressed, the set of healthy buyers shrinks and the potential fire-sale supply rises.

10.16 Figure A2: Total number of aircraft in bankruptcy

How to read. Figure A2 plots the aggregate number of aircraft operated by bankrupt airlines over time. It rises sharply in the early 2000s and peaks during the mid-2000s.

Interpretation. This appendix figure provides the aggregate background behind the model-specific measures. At the peak, a large number of aircraft sit inside bankrupt airlines. Even if only some are sold, lenders have reason to mark down expected collateral recovery values. The figure also helps explain why the paper focuses on the 1994-2007 period: it includes major variation in bankruptcy-induced collateral exposure.

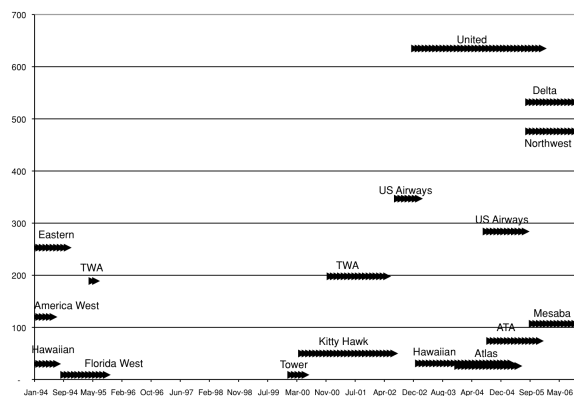


Figure A1. Timeline of airline bankruptcies in the United States, 1994 to 2006. Airline bankruptcy dates are obtained from SDC Platinum.

Figure A1: Timeline of airline bankruptcies in the United States, 1994 to 2006

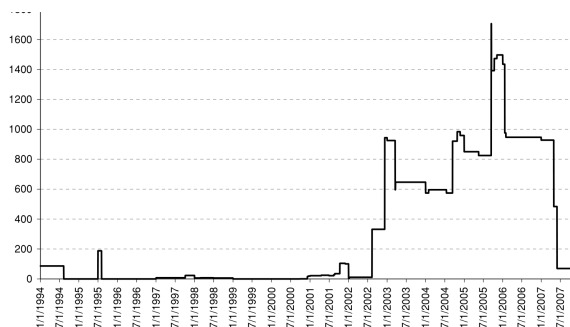


Figure A2. Total number of aircraft in bankruptcy. Total number of aircraft operated by bankrupt airlines in the United States, 1994 to 2007. Airline bankruptcy dates are obtained from SDC Platinum. Fleet data are obtained from the Ascend CASE database.

Figure A2: Total number of aircraft in bankruptcy

11 How to remember the paper

A useful one-sentence summary is: *industry bankruptcies raise solvent competitors' secured borrowing costs when those competitors pledge similar collateral, because bankruptcies reduce the value and redeployability of that collateral.*

The empirical punchline is not just that spreads rise in bad times. The punchline is that spreads rise more for tranches whose exact aircraft collateral overlaps with the fleets of bankrupt airlines, and especially when the claim is junior, high-LTV, or issued by a weak airline. That pattern is what makes the collateral channel interpretation convincing.